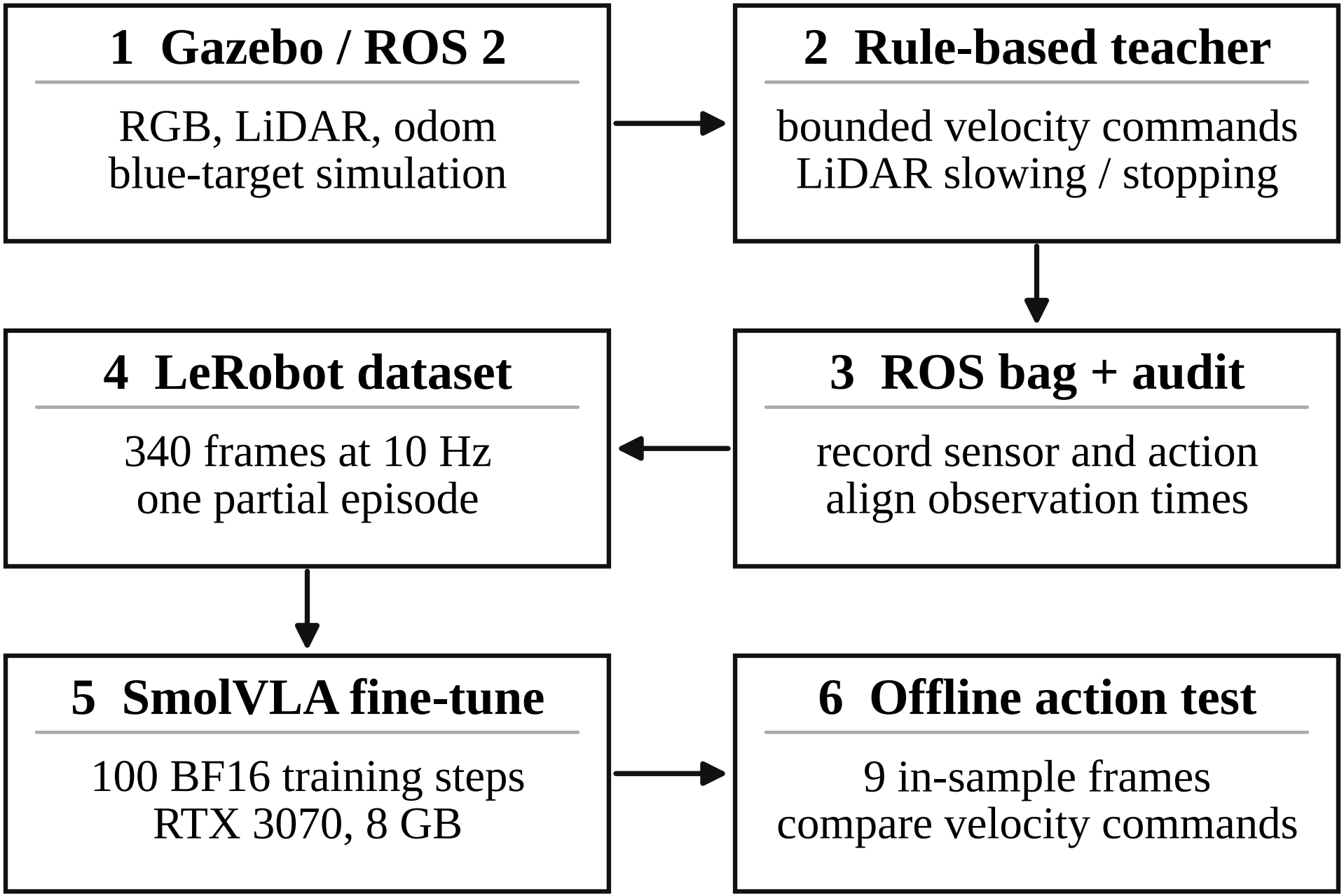
1 Gazebo / ROS 2
RGB, LiDAR, odom
blue-target simulation
2 Rule-based teacher
bounded velocity commands
LiDAR slowing / stopping
3 ROS bag + audit
record sensor and action
align observation times
4 LeRobot dataset
340 frames at 10 Hz
one partial episode
5 SmolVLA fine-tune
100 BF16 training steps
RTX 3070, 8 GB
6 Offline action test
9 in-sample frames
compare velocity commands